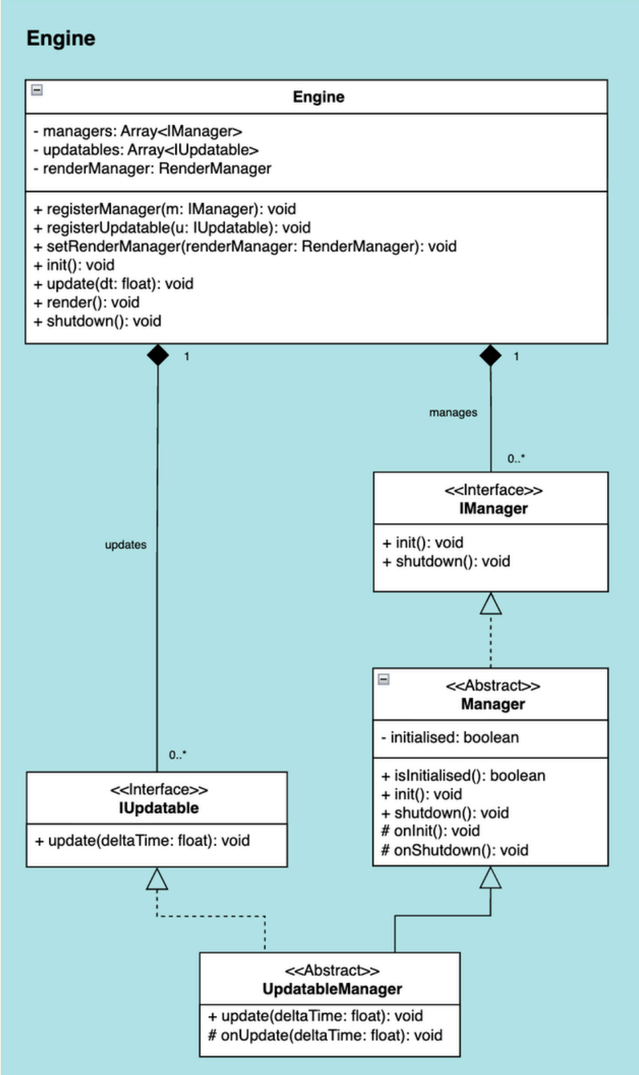
OOP P1_T9 DEMO

Presented by:
Angel Zhang Xin Yu [2503541]
Pham Anh Bao Khang [2503012]
Timothy Chia Kai Lun [2501530]
Jaison Tan Yann Hann [2501161]

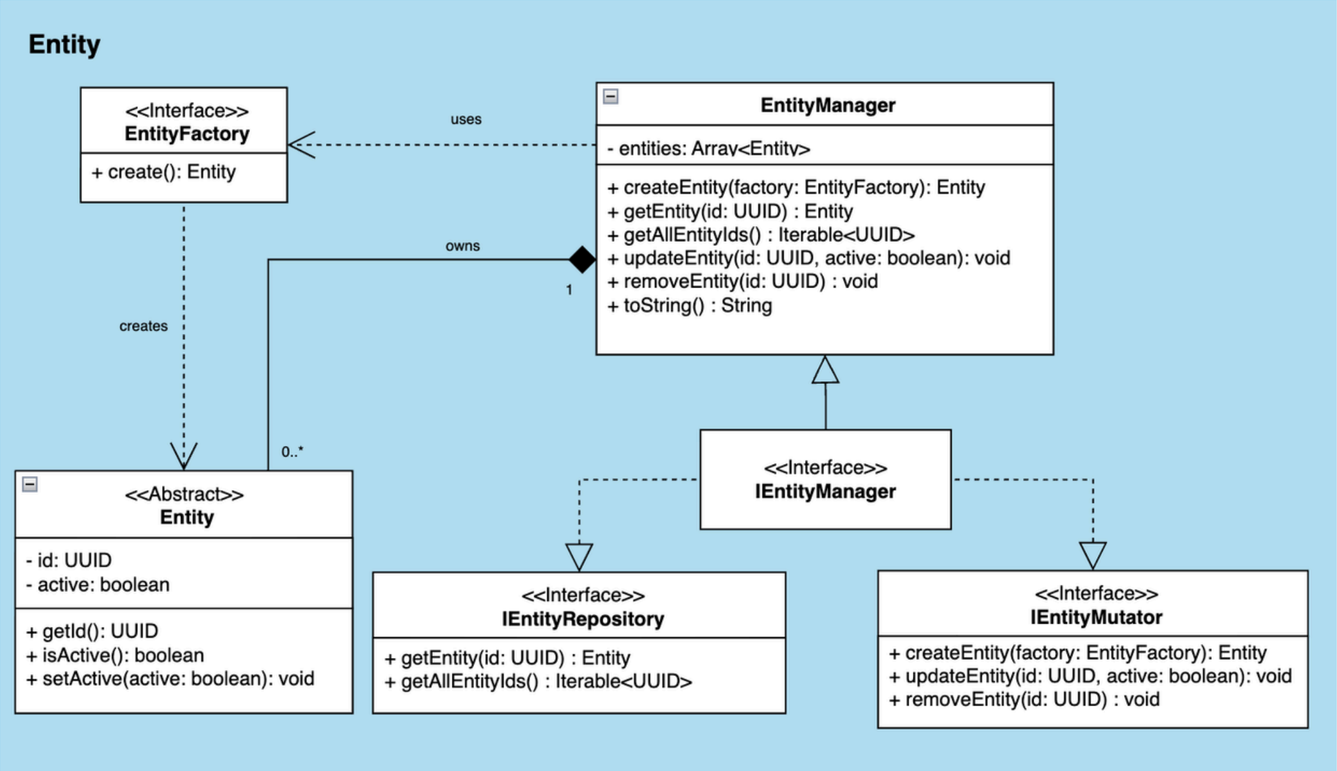
Overview

1. **Engine**
2. **Entity System**
3. **Input System**
4. **Movement System**
5. **Collision System**
6. **Scene System**
7. **Render System**
8. **Final UML Diagram**
9. **Demo**

Engine


Engine
Engine
- managers: Array<IManager>
- updatables: Array<IUpdatable>
- renderManager: RenderManager
+ registerManager(m: IManager): void
+ registerUpdatable(u: IUpdatable): void
+ setRenderManager(renderManager: RenderManager): void
+ init(): void
+ update(dt: float): void
+ render(): void
+ shutdown(): void
1
1
manages
0..*
updates
0..*
<<Interface>>
IManager
+ init(): void
+ shutdown(): void
<<Abstract>>
Manager
- initialised: boolean
+ isInitialised(): boolean
+ init(): void
+ shutdown(): void
# onInit(): void
# onShutdown(): void
<<Interface>>
IUpdatable
+ update(deltaTime: float): void
<<Abstract>>
UpdatableManager
+ update(deltaTime: float): void
# onUpdate(deltaTime: float): void

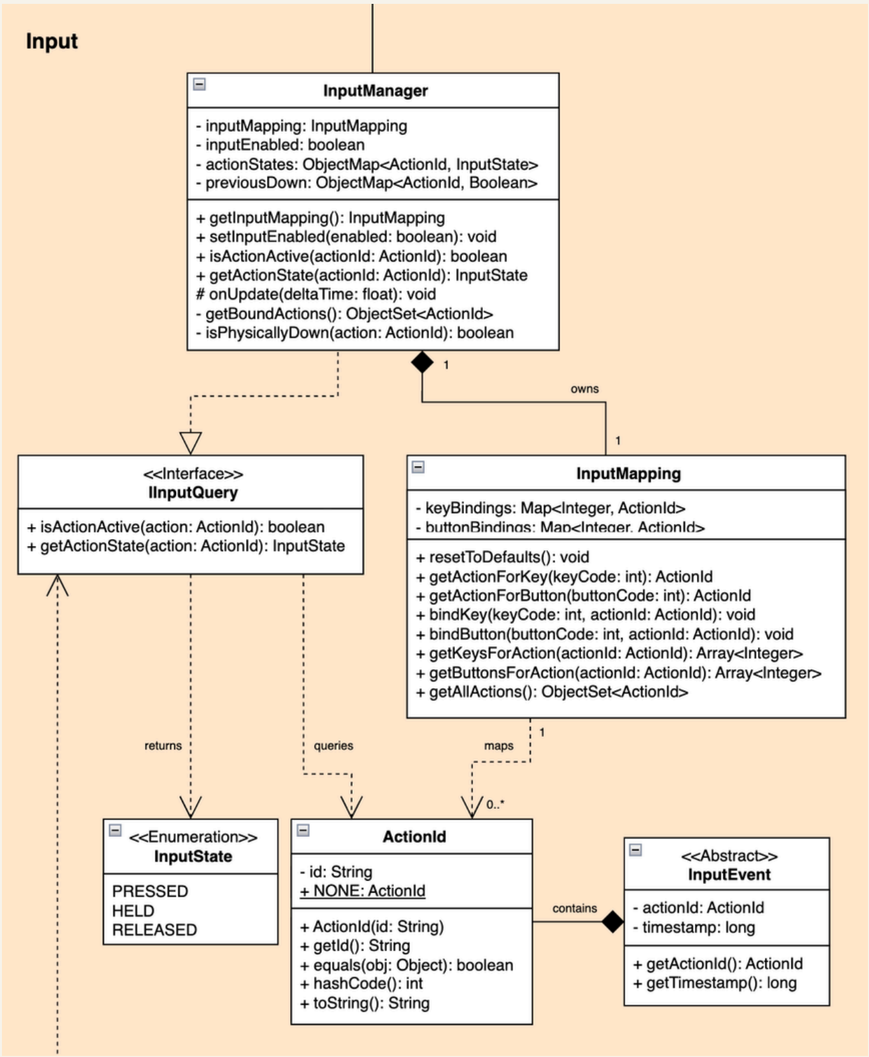
Entity System



EntityManager

- Inherits from abstract Manager class
-

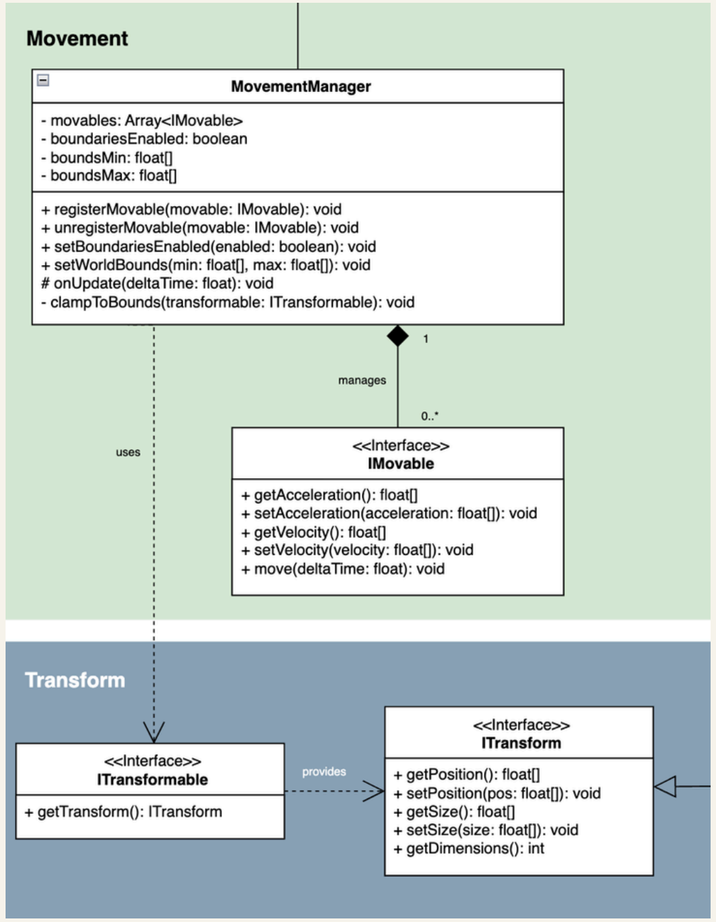
Input System



InputManager

- Inherits from UpdatableManager class
-

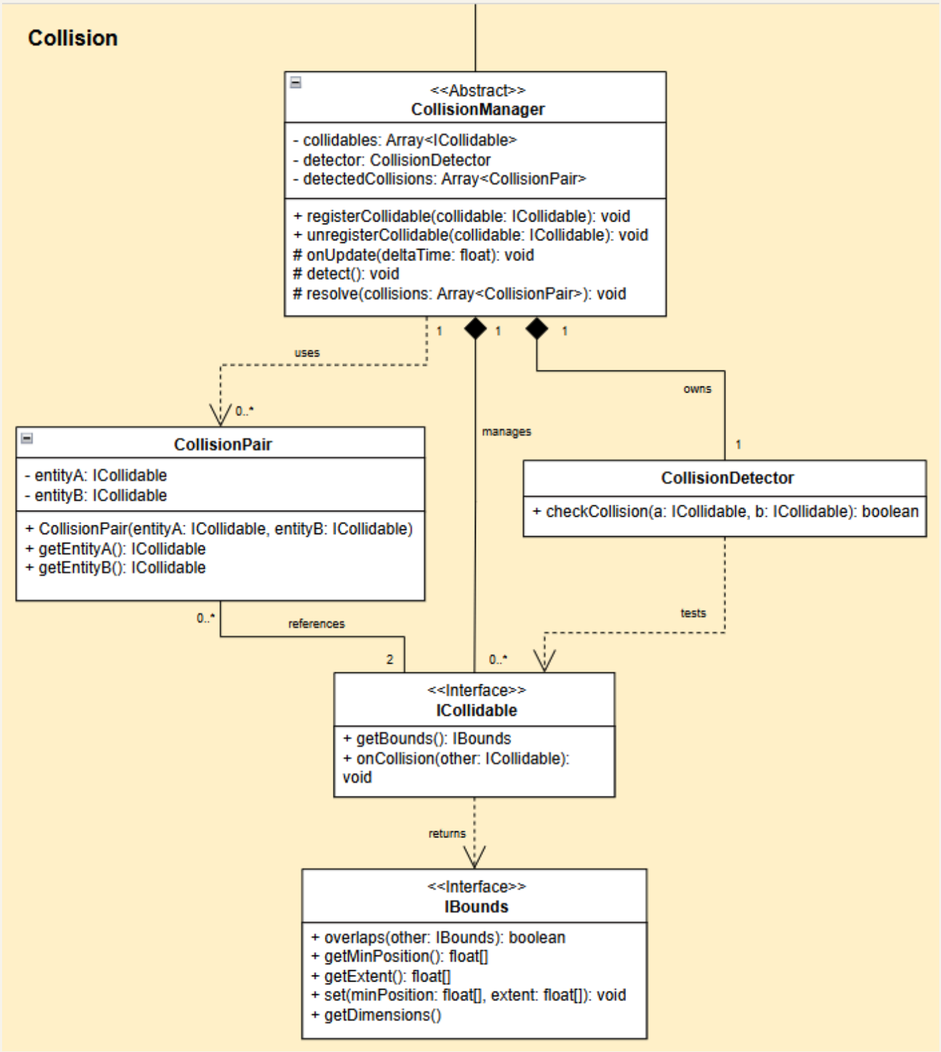
Movement System



MovementManager

- Inherits from UpdatableManager class
-

Collision System



CollisionManager

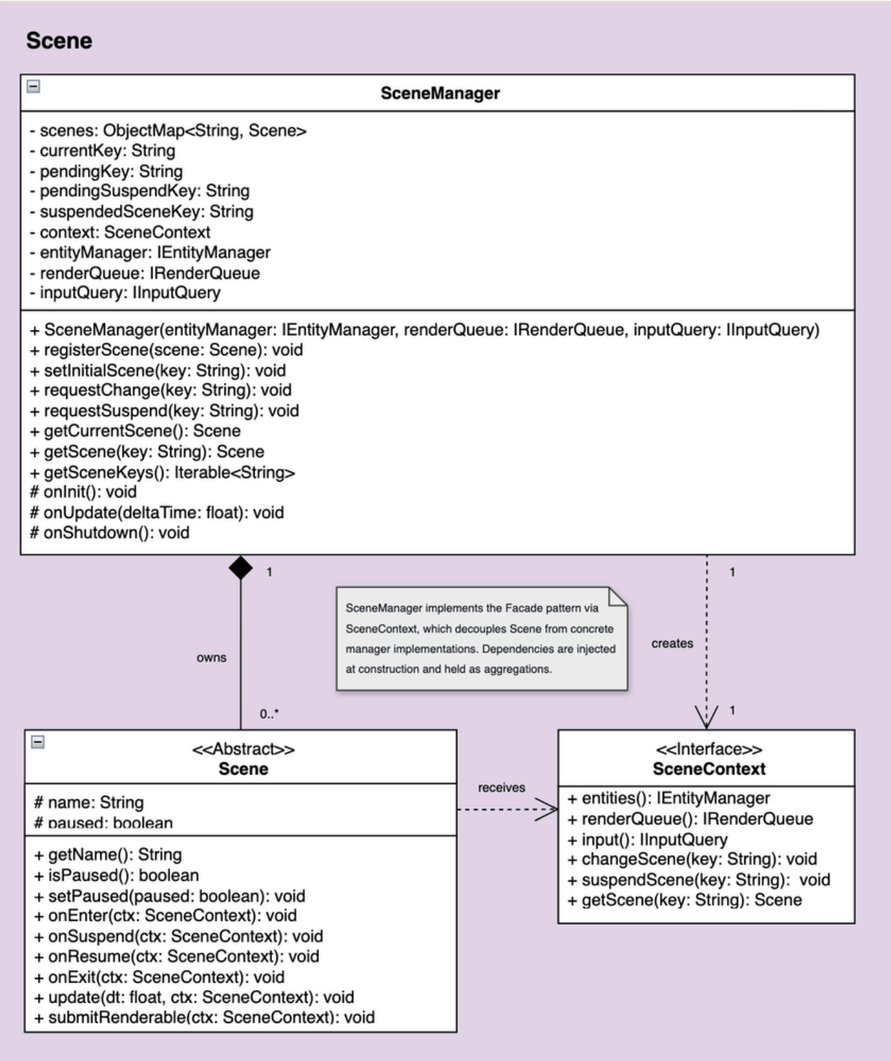
- inherits from UpdatableManager class
- aggregates ICollidable
- composes CollisionDetector
- depends on CollisionPair

CollisionPair references ICollidable

CollisionDetector depends on ICollidable

ICollidable depends on IBounds

Scene System



SceneManager

- Inherits from UpdatableManager class
- Scene Manager composes Scenes
- Scene Manager depends on Scene Context
- Scene depends on Scene Context
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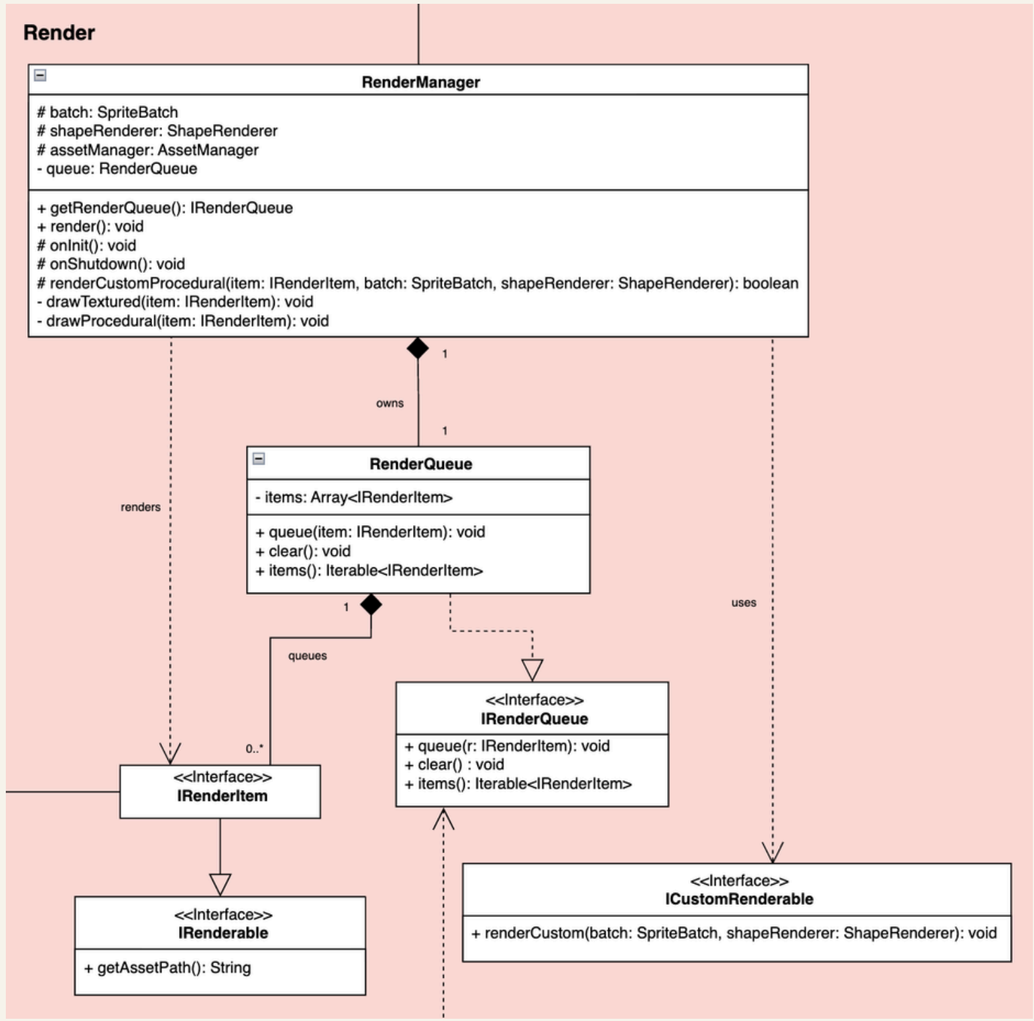
Render System

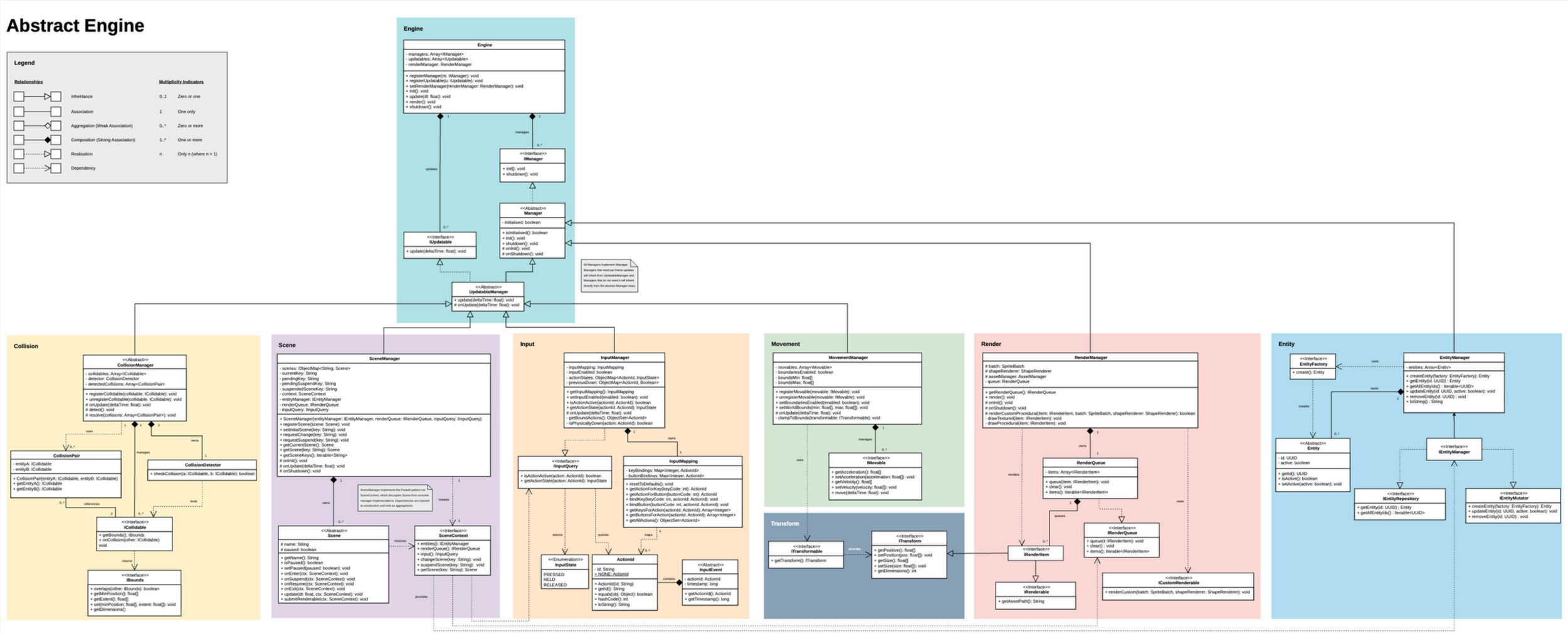


RenderManager

- Inherits from abstract Manager class

Render Manager composes Render Queue
Render Manager depends on iRenderItem
Render Manager depends on iCustomRenderable
Render Queue implements iRenderQueue
Render Queue composes iRenderItem
iRenderItem inherits iRenderable

Final UML Diagram



Demo